INVERSE TRIGONOMETRIC FUNCTIONS

PLANE TRIGONOMETRY

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Domain and Range

Inverse Trigonometric Function

- arcsine
- arccosine
- arctangent

Angle of Elevation and Depression

DOMAIN AND RANGE OF A FUNCTION

DOMAIN

$$x \longrightarrow \boxed{f} \longrightarrow f(x)$$

Set of all valid **inputs** over which the function has defined outputs

$$x \longrightarrow \boxed{\frac{2}{x}} \longrightarrow f(x)$$

All **real numbers** except zero

$\{x \in \boldsymbol{R} | x \neq 0\}$

What is the **domain** of $g(y) = \sqrt{\boldsymbol{y-6}}$?

RANGE

$$x \longrightarrow \boxed{f} \longrightarrow f(x)$$

$$x \longrightarrow \boxed{\frac{2}{x}} \longrightarrow f(x)$$

Set of all possible **outputs**

All **real numbers** except zero

$\{x \in \boldsymbol{R} | x \neq 0\}$

EXERCISE

What is the **domain** and **range** of

$f(x) = \boldsymbol{x^2}$?

INVERSE TRIGONOMETRIC FUNCTIONS

VALUE OF θ

Trigonometric Functions

$$\sin\theta = \frac{opposite}{hyptenuse}$$

$$\cos\theta = \frac{adjacent}{hyptenuse}$$

$$\tan\theta = \frac{opposite}{adjacent}$$

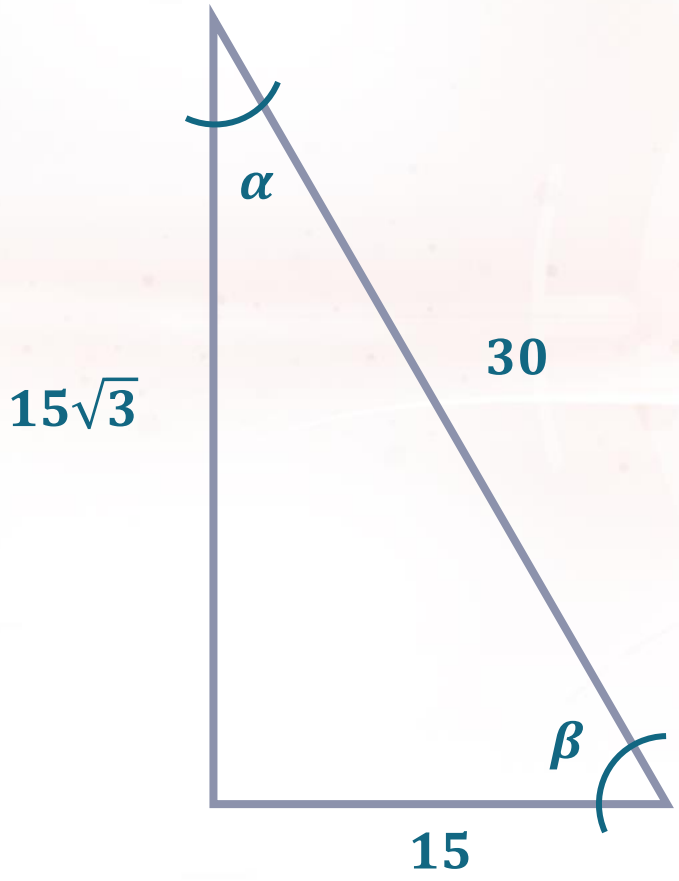
Inverse Trigonometric Functions

$$\theta = \sin^{-1}\left(\frac{opposite}{hyptenuse}\right)$$

$$\theta = \cos^{-1}\left(\frac{adjacent}{hyptenuse}\right)$$

$$\theta = \tan^{-1}\left(\frac{opposite}{adjacent}\right)$$

Determine the **angles** of the given **right-triangle**.

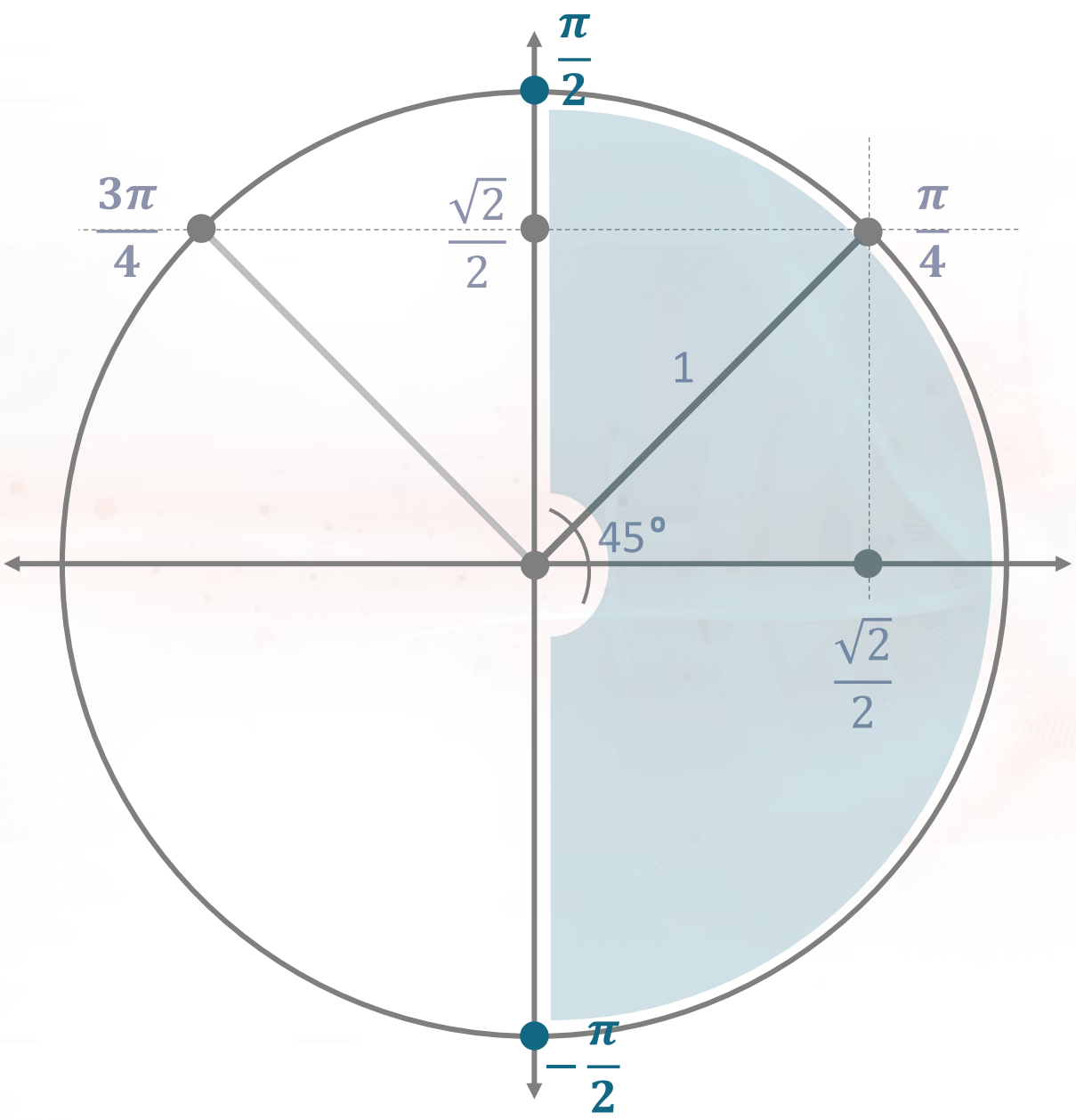


MISCONCEPTION ALERT!

$sin^{-1}\theta$ is **not** the same as $\frac{1}{sin\theta}$

-1 is not an exponent, instead

it simply means **inverse function**



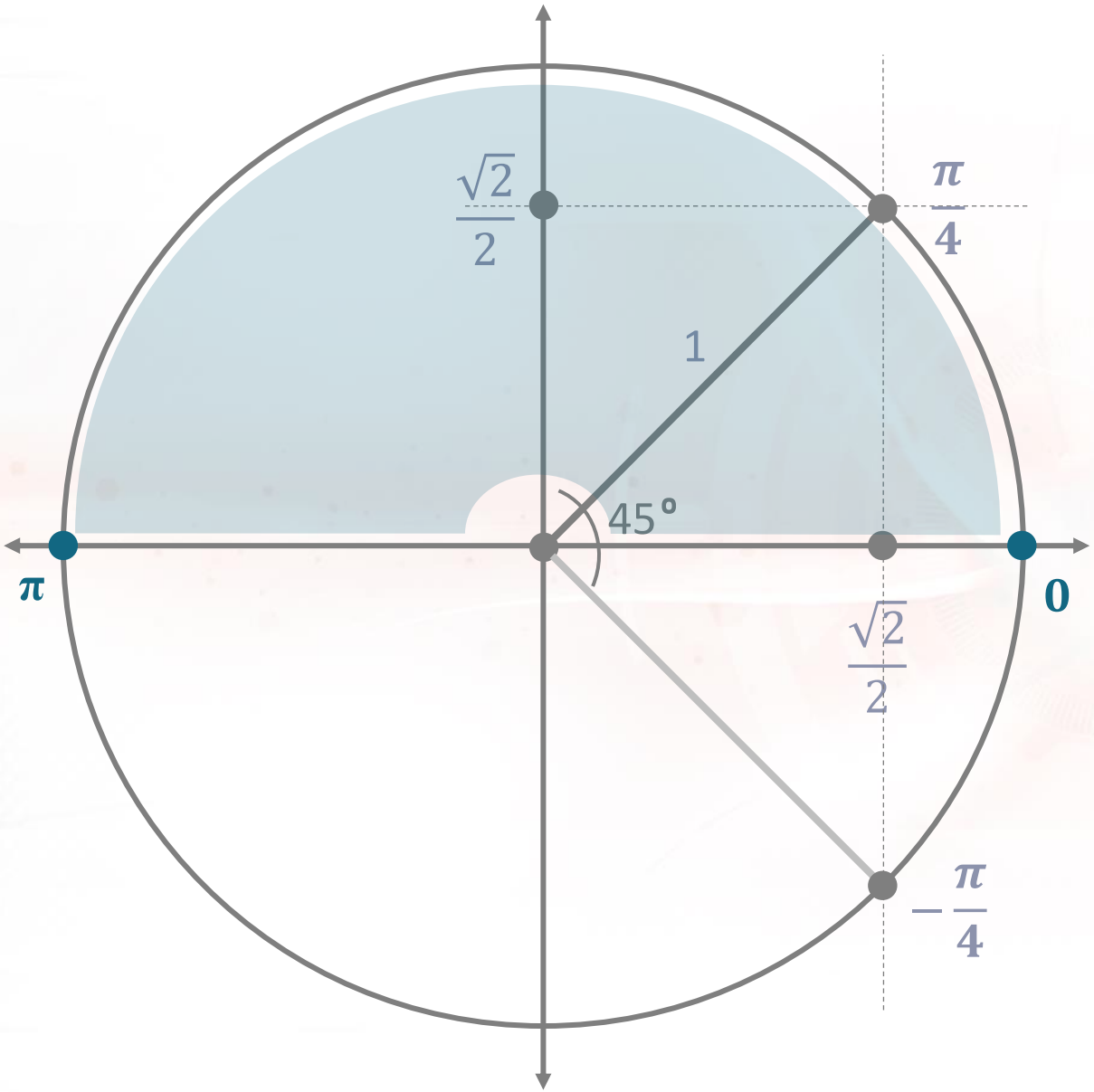
$$x \longrightarrow \boxed{\sin^{-1}x} \longrightarrow \theta$$

Domain:

$$-1 \leq x \leq 1$$

Range:

$$-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$



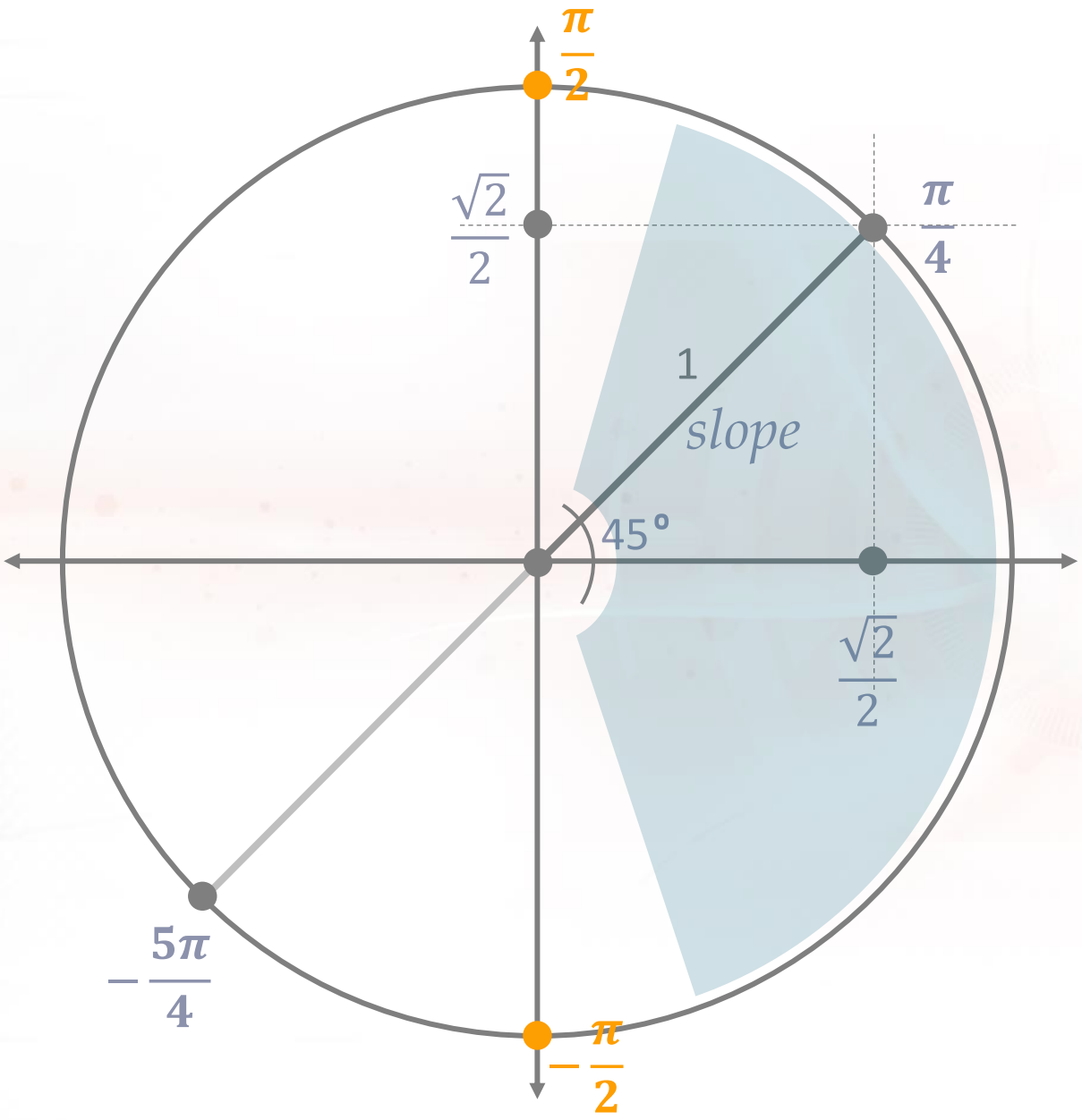
$$x \longrightarrow \boxed{\cos^{-1} x} \longrightarrow \theta$$

Domain:

$$-1 \leq x \leq 1$$

Range:

$$0 \leq \theta \leq \pi$$



$$x \longrightarrow \boxed{tan^{-1}x} \longrightarrow \theta$$

Domain:

$$-\infty < x < \infty$$

Range:

$$-\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

The sine value of all options is 0.98. Which is the **principal** value of **arcsin (0.98)**?

All measures are in radians

A. -11.20

B. -4.91

C. 1.37

D. 7.65

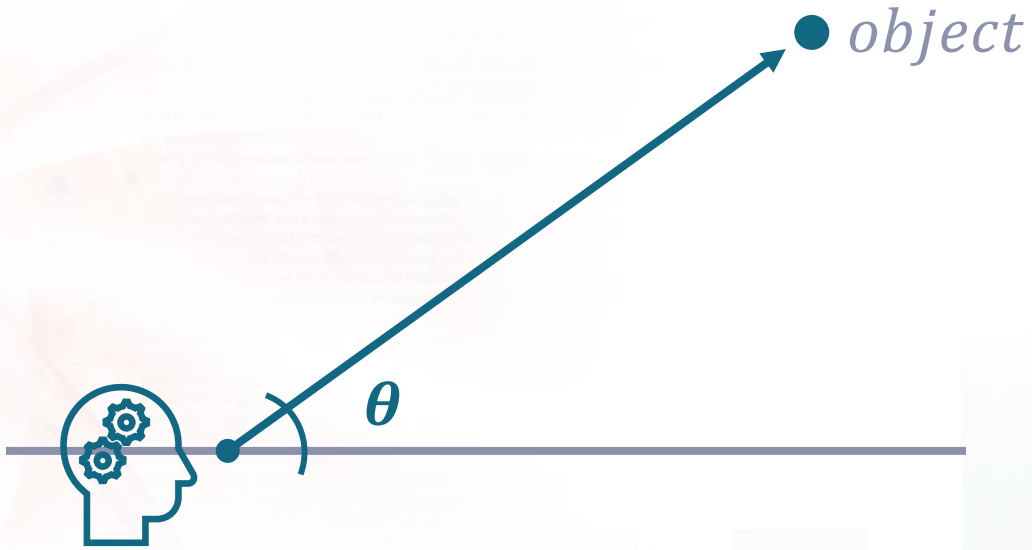
Find the **domain** and **range** of the function

$$f(\theta) = 5sin(\theta) + 4$$

ANGLE OF ELEVATION AND DEPRESSION

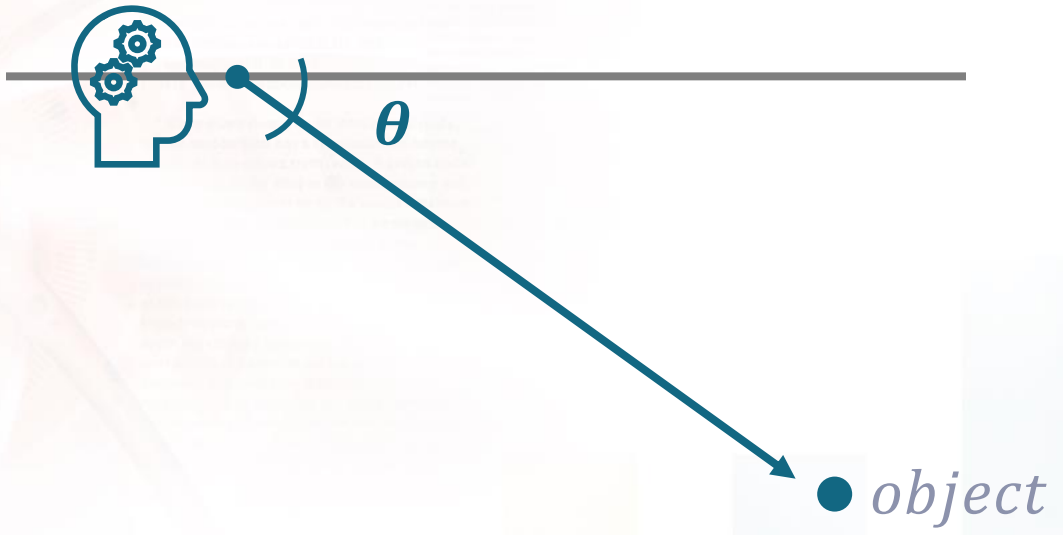
ANGLE OF ELEVATION

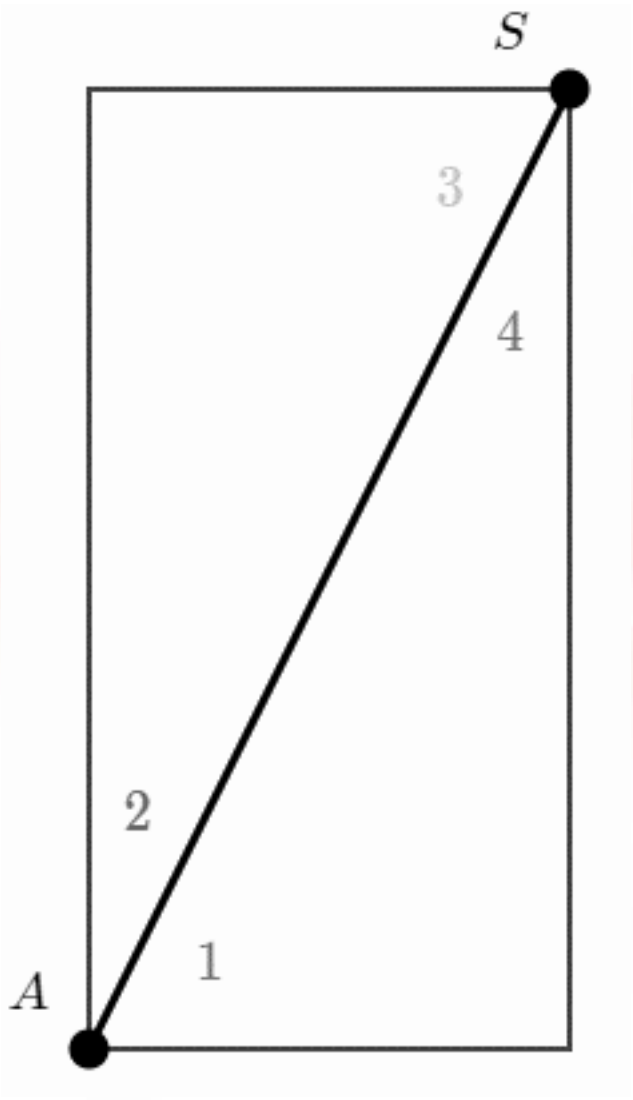
The angle between the horizontal and your line of sight to the object **above** you



ANGLE OF DEPRESSION

The angle between the horizontal and your line of sight to the object **below** you





The given image is a model of Aya, point **A**, **looking up** to Super Girl, point **S**. What is the angle of **elevation** from Aya to Super Girl?

A. $\angle 1$

B. $\angle 2$

C. $\angle 3$

D. $\angle 4$

What is the angle of **depression** from Super Girl to Aya?

You are watching a hot-air balloon that was **300** feet **from you** when it started rising from the ground. What is the angle of **elevation** when the hot-air balloon is **500** feet **above** the ground?

EXERCISE

A tree is struck by lightning and snaps off **34** feet **above** the ground. The top part of the tree, **117** feet **long**, rests with the tip on the ground, the broken end rests on the top of the stump. What **angle** does the top part of the tree make with the ground?

SEATWORK